Appendix to the ETCS Framework

ETCS (Emergency Only Time sequence Causal information System) Based Architecture for Autonomous Driving

(ETCS: refer to the proposal section presented in the main paper)

Excerpt from the paper Explainability and Accountability in End-to-End Autonomous Driving under Real-World Long-Tail Events, first published on December 7, 2025.

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The four papers to be compared with the main paper are as follows:

[1] A. Kuznietsov et al., "Explainable AI for Safe and Trustworthy Autonomous Driving: A Systematic Review," IEEE Trans. Intell. Transp. Syst., vol. 25, no. 12, pp. 19342–19364, 2024.

[2] S. Tekkesinoglu, A. Habibovic, and L. Kunze, "Advancing Explainable Autonomous Vehicle Systems: A Comprehensive Review and Research Roadmap," ACM Trans. Hum.-Robot Interact., vol. 14, no. 3, 2025.

[3] M. Sanchi, "Systemic Explainability for Autonomous Driving: Connecting Ethics to Law and Society," AI & Society, 2026.

[4] S. Atakishiyev, M. Salameh, H. Yao, and R. Goebel, "Explainable Artificial Intelligence for Autonomous Driving: A Comprehensive Overview and Field Guide for Future Research Directions," IEEE Access, vol. 12, pp. 101603–101625, 2024.

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Assessment of four papers across five criteria central to ETCS (criteria: Explainability, Causality, Counterfactuals, Logging, and ODD Constraints)

1. **Definition of Explainability**

ETCS in the main paper understands explainability not as the mere visualization of behavioral logs, but as a concept that encompasses purpose hierarchies, value priorities, and the agents responsible for determining those priorities. By linking Pearl's three-layer causal model with hierarchical purposes, it seeks to explain not only what a system did, but also why particular purposes and priorities were selected.

The IEEE XAI safety review [1] examines explainability in relation to safe and trustworthy autonomous driving and proposes the SafeX framework. ACM [2] organizes explainability around explanatory tasks, explanatory information, and the communication of explanations for different stakeholders. Springer [3] develops the concept of "systemic explainability," connecting technical explanations with ethics, law,

and society. Atakishiyev et al. [4] survey XAI methods for autonomous driving and propose a conceptual framework for explainable end-to-end systems, emphasizing timely explanations, transparency, regulatory compliance, and societal acceptance. While these studies address safety, stakeholder needs, and system-level explainability, ETCS places particular emphasis on explicitly linking purpose hierarchies, value priorities, and the agents responsible for establishing them.

2. **Causal Modeling of Purpose and Task**

ETCS distinguishes Purpose, represented through objective functions, constraints, and evaluation metrics, from Task, treated as a concrete intervention. It models these as distinct but hierarchically connected layers in a causal framework, enabling analysis of how objectives and the actions used to achieve them become entangled in end-to-end (E2E) systems.

The IEEE XAI safety review [1] integrates explainability with safety assurance through the SafeX framework. ACM [2] structures its analysis around explanation tasks, information, and communication. Springer [3] examines system-level explainability and its implications for accountability across technical, legal, and social contexts. Atakishiyev et al. [4] propose a conceptual framework for explainable E2E autonomous driving and discuss the methods and components needed to explain driving decisions. However, their framework is primarily concerned with the generation, timing, and delivery of explanations, whereas ETCS explicitly separates Purpose from Task and makes their hierarchical causal relationship the central focus of its model.

3. **Counterfactual Reasoning and Unselected Tasks**

ETCS explicitly treats counterfactual evaluation of unselected tasks as a basis for examining alternative decisions and allocating responsibility. It proposes combining high-density logging triggered by emergencies with post-event counterfactual analysis on the cloud.

The IEEE XAI safety review [1] surveys counterfactual reasoning and other XAI methods for safe autonomous driving, but its proposed SafeX framework focuses on integrating explanation methods with safety assurance rather than specifying an emergency-triggered logging and post-event counterfactual-analysis architecture. ACM [2] examines explanation requirements for autonomous-driving stakeholders, but a comparable operational framework for evaluating unselected tasks is not established in the aspects discussed here. Springer [3] emphasizes causal reconstruction, accountability, and responsibility attribution across the autonomous-driving lifecycle, primarily from a systemic, ethical, and legal perspective. Atakishiyev et al. [4] explicitly

review counterfactual explanations for driving decisions and propose a conceptual framework for explainable E2E driving; however, their treatment does not specify the same combined architecture of emergency-triggered high-density logging and subsequent counterfactual evaluation of unselected tasks as ETCS.

4. **Specificity of Logging Design**

ETCS proposes a concrete logging scheme that records TTC (time to collision), Time Budget, purpose-state IDs, purpose-switching timestamps, candidate task sets, and evaluation distributions at high temporal resolution during emergencies. These records are intended to support subsequent counterfactual analysis and responsibility assessment.

The IEEE XAI safety review [1] proposes the SafeX framework for integrating explanation delivery and safety assurance, but does not specify this same set of logging fields and emergency-trigger conditions. ACM [2] addresses the information required to generate and communicate explanations, but does not present an equivalent emergency-triggered logging specification in its principal framework. Springer [3] discusses explainability as a means of supporting causal reconstruction and accountability, but does not set out the same detailed technical logging scheme. Atakishiyev et al. [4] present a conceptual framework for explainable E2E autonomous driving and review counterfactual methods, but do not specify ETCS's particular combination of logging fields, emergency-trigger conditions, and high-density recording for post-event analysis.

5. **ODD, Physical Constraints, and Causal Stability**

ETCS explicitly proposes a strategy for stabilizing causal inference by deliberately restricting physical degrees of freedom through speed limits, inter-vehicle distance requirements, restrictions on abrupt maneuvers, and ODD narrowing under nighttime or adverse weather conditions.

The IEEE XAI safety review [1] addresses ODDs and safety requirements, but the integration of physical constraints from the perspective of causal inference stability is less emphasized. ACM [2] recognizes the challenges posed by ODDs and the expansion of driving scenarios, but does not integrate physical constraints with causal stability. Springer [3] discusses ODD restrictions from an institutional perspective, while their technical integration with causal inference remains abstract. Atakishiyev et al. [4] also address ODDs and safety-oriented design, but do not specify a constraint strategy for causal stability as concretely as ETCS.

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Postscript ＊＊＊＊Software and the Essence of 'New Value' Creation＊＊＊＊

A few illustrative episodes for the purpose of considering the limits of hardware performance and the possibility of redefining value

In the film Back to the Future, there are two technically striking scenes that suggest a shift in how the value of physical objects can change, not through "physical performance," but through a change in perception.

The first is the "Enchantment Under the Sea" school dance party. Marty borrows a professional guitarist's instrument, plays it in a self-absorbed frenzy, and leaves the audience bewildered. When the guitarist gets his instrument back, he stares at it as if wondering, "Is this really my guitar?" The second is the chase scene in which Marty, pursued by Biff's car, borrows a child's small wooden scooter. Marty instantly transforms it into a skateboard and uses its characteristics to escape. When he returns the board, the children look at it in astonishment, as if questioning whether this object is truly the toy they had been playing with.

These two episodes suggest that the claim that "Software cannot surpass the physical limits of hardware" implicitly presupposes a fixed and immutable set of hardware performance metrics. Thus, this claim assumes that the existing yardsticks for measuring hardware performance remain permanently valid. Yet even if physical performance ceilings do not change, software can alter environmental perception, action selection, and functional integration, thereby changing the roles and value that hardware can fulfill. With the same sensors and actuators, different software can expand the range of "what can be done." This is not performance improvement; it is a redefinition of the very concept of performance.

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A real historical episode from the development of electronic fuel injection (EFI) systems in the 1960s illustrates a different kind of change in technical value.

Engine development was typically divided between the main engine group (responsible for the cylinder block, etc.) and the auxiliary group (responsible for carburetors and other peripherals). During the EFI project, engineers in the main engine group expressed dismissive views, stating that EFI does not improve the engine's core performance. In response, the EFI engineers said: "Indeed, EFI may not directly enhance the engine's inherent performance. However, even drivers who are not accustomed to vehicle operation can benefit from EFI's precise air–fuel ratio control, which optimizes fuel injection across all driving conditions. As a result, overall fuel economy is improved. The value of EFI lies in its capacity to allow non-professional drivers to extract a higher level of the inherent performance of this engine, which is a

hardware system." Here, the value created by EFI was not an increase in the engine's physical hardware performance. It was the emergence of a 'new value' system in which anyone could extract high performance from the engine.

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The conclusion is that evaluating physical AI should not hinge on whether it can break hardware limits. Rather, the question is whether software can use existing hardware to realize possibilities that conventional evaluation metrics never anticipated, changing the definitions of performance and value themselves.

Scoring higher on an existing scale and updating the scale itself are fundamentally different. Marty's guitar performance and his reinterpretation of the wooden scooter as a skateboard symbolize precisely this distinction. Likewise, the EFI case shows that technology can transform functions, user accessible performance, and ultimately value, without altering the underlying hardware's physical capabilities.

The essence of the 'new value' that software aim to create is not the pursuit of surpassing physical hardware limits. It is the expansion of human value recognition by changing the boundaries of existing value itself.

With kindest regards,

Akira.

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